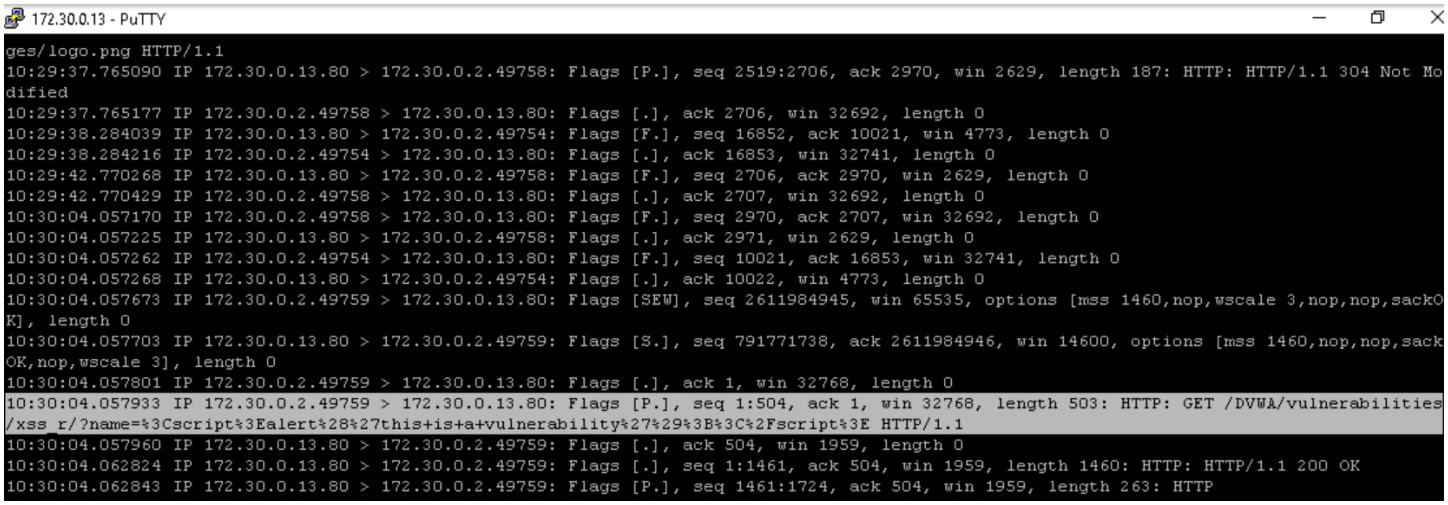
Lab 7: Analyzing Network Traffic to Create a Baseline Definition

In this lab, we used various techniques to identify attacks and capture information about "suspicious" file transfers. Using TCPdump, we were able to identify an XSS attack by searching the logs produced during the time of the attack. In addition, we were able to use Wireshark to capture FTP and TFTP packets while files were being transferred between devices. We were able to analyze this "suspicious" file transfer activity using NetWitness and access information like IP addresses, file names, and passwords used.

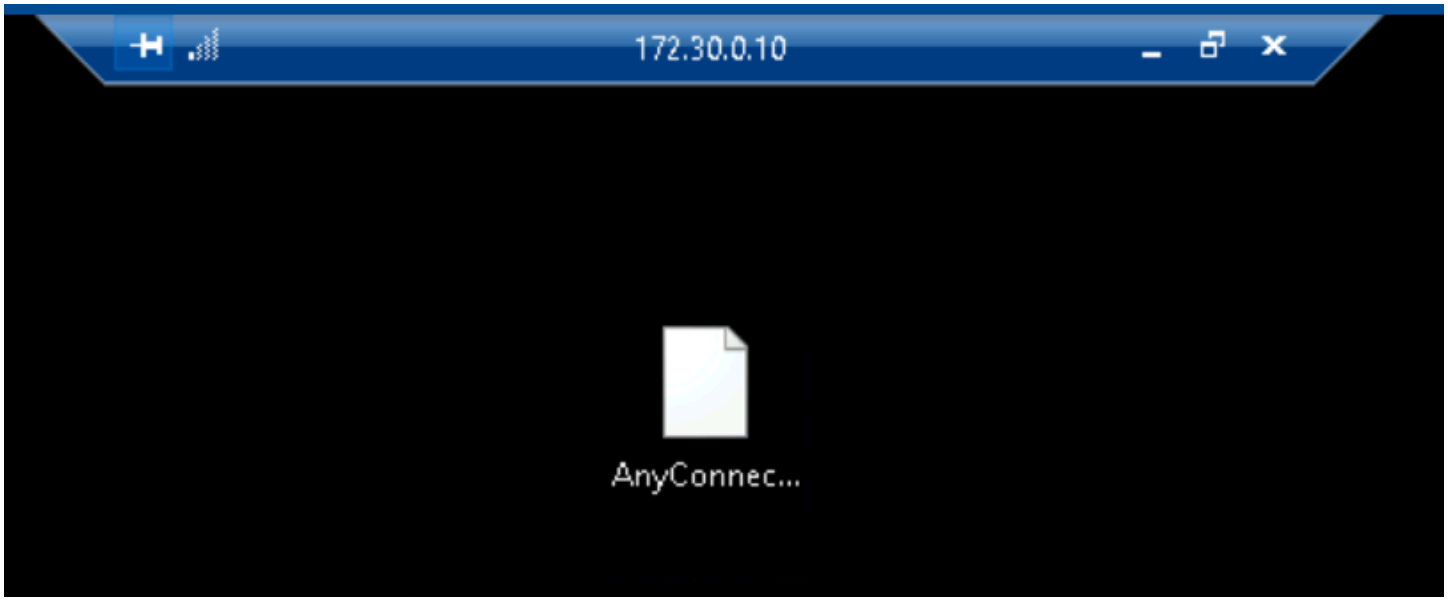
SECTION 1

Part 1: Capture Network Traffic Using TCPdump Utility

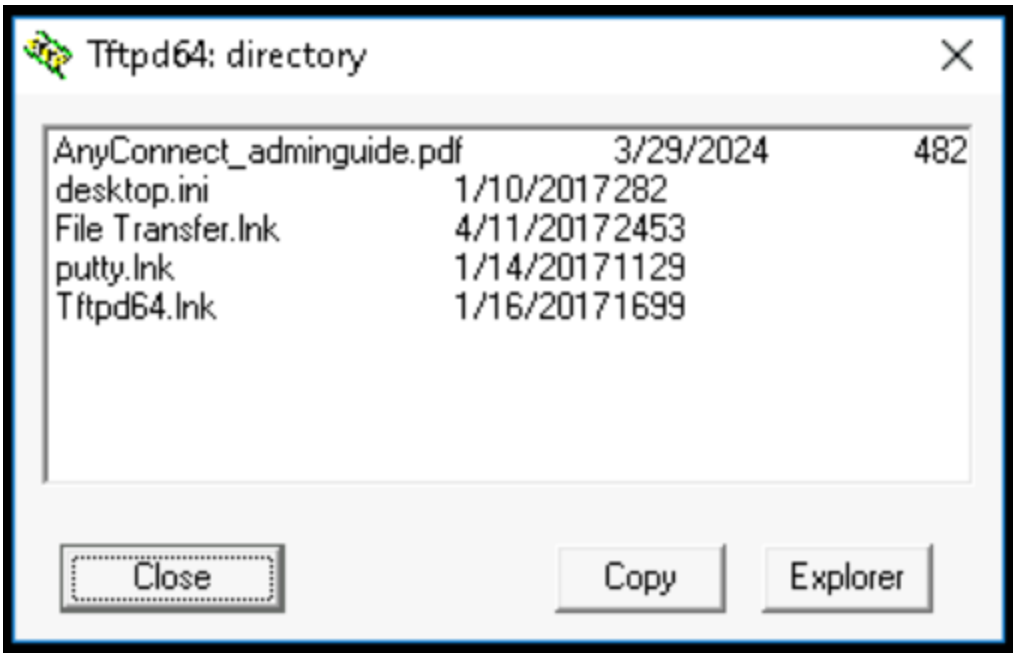


Screenshot 1: The alert script ran showing the XSS vulnerability during the TCPdump scan

Part 3: Transfer Files using Tftpd64 and FileZilla



Screenshot 2: PDF file successfully transferred to the TargetWindows02 desktop with FileZilla



Screenshot 3: PDF file successfully transferred to vWorkstation with Tftpd64

Part 4: Analyze a Packet Capture with Wireshark

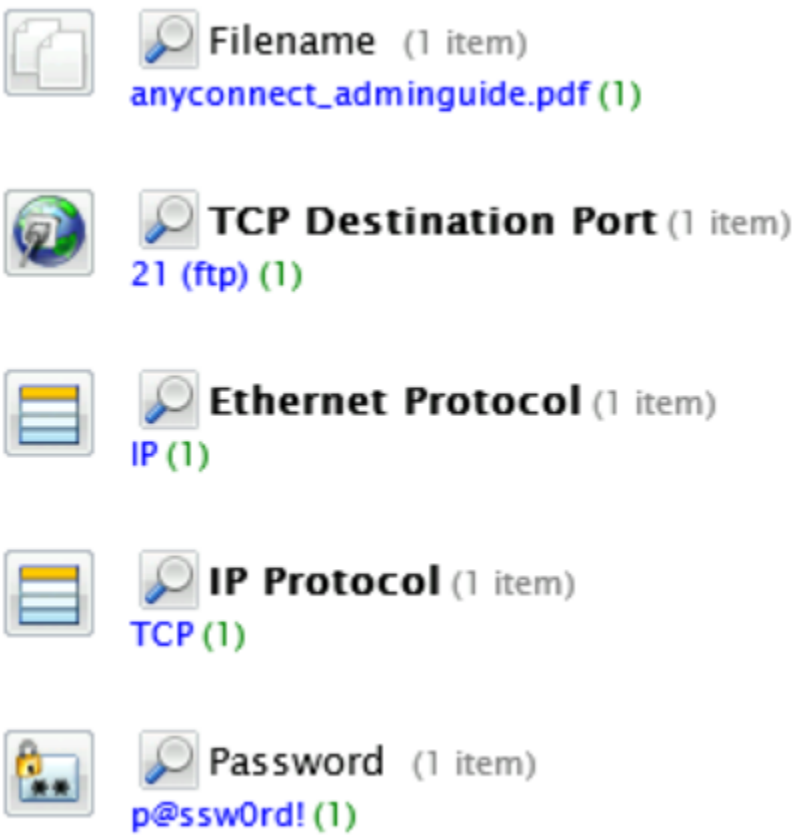
Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets
Frame	100.0	21232	100.0	8384423	607 k	0
Ethernet	100.0	21232	3.5	297248	21 k	0
Internet Protocol Version 4	100.0	21232	5.1	424640	30 k	0
User Datagram Protocol	88.8	18861	1.8	150888	10 k	0
Trivial File Transfer Protocol	88.8	18858	58.5	4902354	355 k	9430
Data	44.4	9428	57.6	4826866	349 k	9428
NetBIOS Name Service	0.0	3	0.0	150	10	3
Transmission Control Protocol	11.2	2371	29.5	2477063	179 k	662
FTP Data	7.9	1667	29.0	2428156	175 k	1667
File Transfer Protocol (FTP)	0.2	42	0.0	1391	100	42

Screenshot 4: Protocol hierarchy tab in Wireshark after FTP packet capture

Topic / Item	Count	Average	Min val	Max val	Rate (ms)	Percent	Burst rate	Burst start
Packet Lengths	21232	394.90	54	1514	0.1922	100%	24.7700	110.240
0-19	0	-	-	-	0.0000	0.00%	-	-
20-39	0	-	-	-	0.0000	0.00%	-	-
40-79	10113	59.64	54	73	0.0915	47.63%	12.3800	110.240
80-159	21	96.95	85	138	0.0002	0.10%	0.0800	0.000
160-319	5	230.40	176	294	0.0000	0.02%	0.0300	0.000
320-639	9427	558.00	558	558	0.0853	44.40%	12.3900	110.240
640-1279	7	879.14	858	1006	0.0001	0.03%	0.0700	15.660
1280-2559	1659	1514.00	1514	1514	0.0150	7.81%	16.5900	15.648
2560-5119	0	-	-	-	0.0000	0.00%	-	-
5120 and greater	0	-	-	-	0.0000	0.00%	-	-

Screenshot 5: Packet lengths tab in Wireshark after FTP packet capture

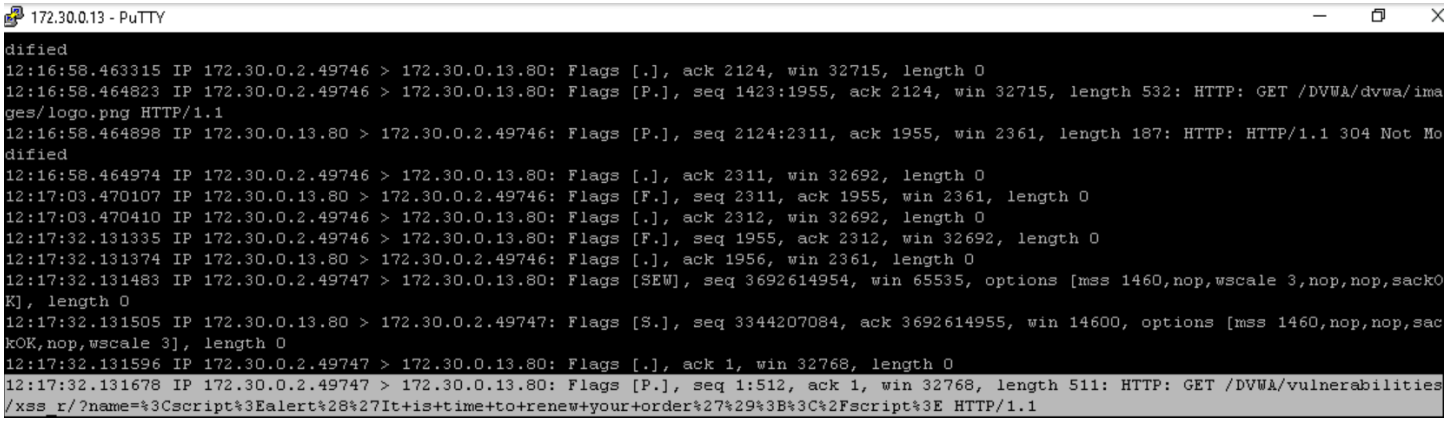
Part 5: Analyze a Packet Capture File with NetWitness Investigator



Screenshot 6: The file name and password captured by Wireshark during the previous lab activities

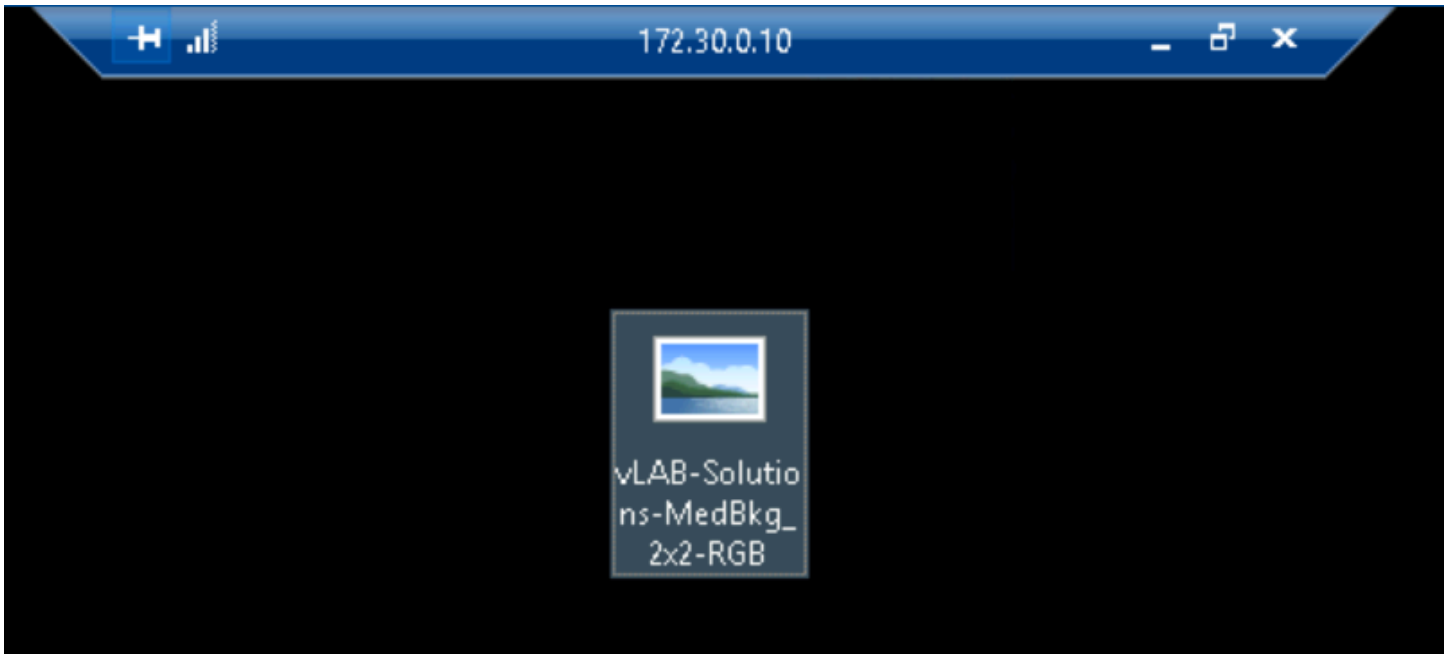
SECTION 2

Part 1: Capture Network Traffic Using TCPdump Utility

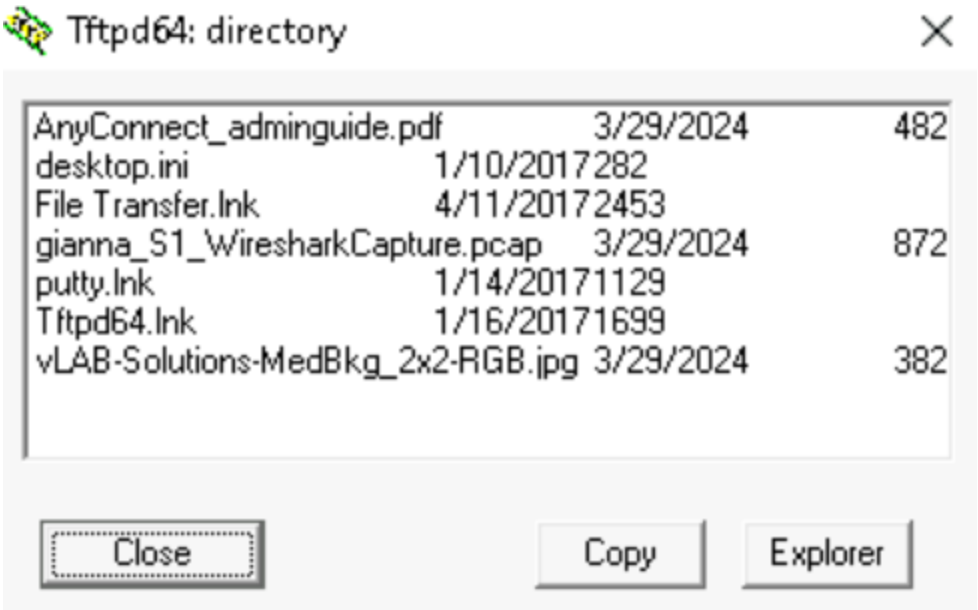


Screenshot 7: The alert script showing the XSS vulnerability DVWA

Part 3: Transfer Files using Tftpd64 and FileZilla



Screenshot 8: JPG file successfully transferred to the TargetWindows02 desktop with FileZilla



Screenshot 9: JPG file successfully transferred to vWorkstation with Tftpd64

Part 4: Analyze a Packet Capture with Wireshark

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s
Frame	100.0	1233	100.0	211619	3172	0	0	0
Ethernet	100.0	1233	8.2	17262	258	0	0	0
Internet Protocol Version 4	98.4	1213	11.5	24260	363	0	0	0
User Datagram Protocol	12.4	153	0.6	1224	18	0	0	0
Trivial File Transfer Protocol	12.3	152	18.4	38892	582	77	368	5
Data	6.1	75	18.1	38224	572	75	38224	572
NetBIOS Datagram Service	0.1	1	0.1	201	3	0	0	0
SMB (Server Message Block Protocol)	0.1	1	0.1	119	1	0	0	0
SMB MailSlot Protocol	0.1	1	0.0	25	0	0	0	0
Microsoft Windows Browser Protocol	0.1	1	0.0	33	0	1	33	0
Transmission Control Protocol	86.0	1060	59.9	126827	1901	397	8204	122
Telnet	29.4	362	15.7	33197	497	362	33197	497
SSH Protocol	18.2	224	15.2	32101	481	224	32101	481
FTP Data	2.4	29	18.4	38941	583	29	38941	583
File Transfer Protocol (FTP)	3.9	48	0.8	1592	23	48	1592	23
Address Resolution Protocol	1.6	20	0.3	560	8	20	560	8

Screenshot 10: Protocol hierarchy tab in Wireshark after FTP packet capture

Topic / Item	Count	Average	Min val	Max val	Rate (ms)	Percent	Burst rate	Burst start
Packet Lengths	1233	171.63	42	1514	0.0023	100%	1.8100	222.243
0-19	0	-	-	-	0.0000	0.00%	-	-
20-39	0	-	-	-	0.0000	0.00%	-	-
40-79	842	57.63	42	79	0.0016	68.29%	0.7600	533.694
80-159	184	120.66	81	150	0.0003	14.92%	0.8900	222.243
160-319	48	193.65	166	318	0.0001	3.89%	0.2700	222.253
320-639	104	529.59	321	630	0.0002	8.43%	0.7500	533.694
640-1279	12	951.58	672	1158	0.0000	0.97%	0.0300	278.282
1280-2559	43	1514.00	1514	1514	0.0001	3.49%	0.2600	343.964
2560-5119	0	-	-	-	0.0000	0.00%	-	-
5120 and greater	0	-	-	-	0.0000	0.00%	-	-

Screenshot 11: Packet Lengths tab in Wireshark after FTP packet capture

File

Name:	C:\Users\ADMINI~1\AppData\Local\Temp \2\wireshark_414269F4-8DF0-4852-8F56-384E405060F3_20240329172308_a03840.pcapng
Length:	252 kB
Format:	Wireshark/... - pcapng
Encapsulation:	Ethernet

Time

First packet:	2024-03-29 17:23:14
Last packet:	2024-03-29 17:32:07
Elapsed:	00:08:53

Capture

Hardware:	Unknown
OS:	64-bit Windows Server 2016, build 14393
Application:	Dumpcap (Wireshark) 2.2.3 (v2.2.3-0-g57531cd)

Interfaces

Interface	Dropped packets	Capture filter	Link type	Packet size limit
\Device \NPF_{414269F4-8DF0-4852-8F56-384E405060F3}	0 (0 %)	not port 3389	Ethernet	262144 bytes

Statistics

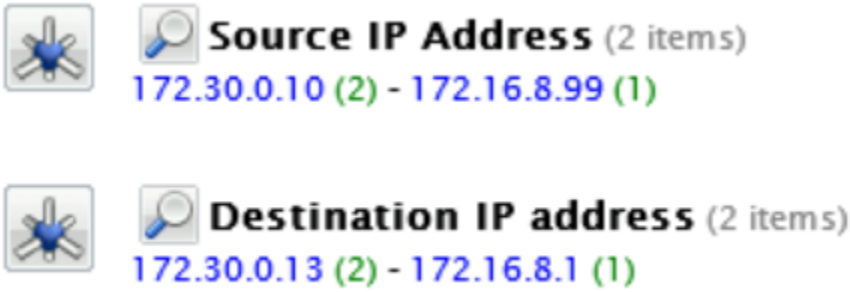
Measurement	Captured	Displayed	Marked
Packets	1233	1233 (100.0%)	N/A
Time span, s	533.706	533.706	N/A
Average pps	2.3	2.3	N/A
Average packet size, B	171.5	171.5	N/A

Screenshot 12: Capture file properties tab in Wireshark after FTP packet capture

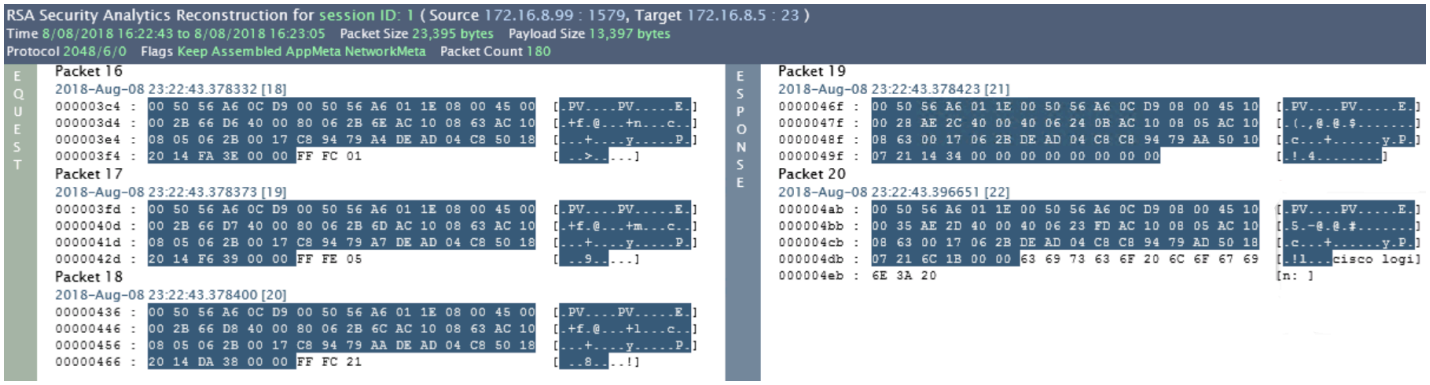
Part 5: Analyze a Packet Capture File with NetWitness Investigator

Screenshot 13: The file name and password captured by Wireshark during FTP activity

Screenshot 14: The file name and password captured by Wireshark during TFTP activity



Screenshot 15: The file name and password captured by Wireshark during SSH activity



Screenshot 16: Side by Side view of search for “cisco” in packet capture